The multiplicity-m leading term for general ξ, η

Claude, with Daniel Murfet

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Abstract

Combining the freezing limit (unit 78) with the frozen coefficient theorem (unit 75): for a block of $m = d_0 + 2$ coordinates with common exponent $p = (h_i + 1)/k_i$, noncritical coordinates with exponents $q_i' > p$, and jointly continuous ξ, η,

$$\frac{N^p\, Z(N)}{\log^{d_0+1} N} \longrightarrow C(\xi, \eta) = \frac{\prod_i k_i^{-1}}{(d_0+1)!} \int_{(0,b]^{d'}} v^{h'} (v^{k'})^{-p} A_{p-1}(\xi(0, v))\, \eta(0, v)\, dv,$$

with $C > 0$ when $\eta(0, \cdot) > 0$; hence $Z(N) \sim C\, N^{-p} \log^{d_0+1} N$ and, at $N = \sqrt{n}$, $Z \sim (C/2^{d_0+1})\, n^{-p/2} \log^{d_0+1} n$. This is the leading term of `thm:TaylorTree` for general continuous ξ, η and multiplicity $m \geq 2$; Programme A (unit 72) covers $m = 1$. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem integral_blockDivisorDensity_pos ...
    (hepos :   v, (  i, 0 < v i   v i   b) → 0 < e v) :
    0 <   v, blockDivisorDensity   p k d k' h' a e v  (boxMeasure b d')
noncomputable def blockCoeff (  b p :  ) {d  d' :  } (k : Fin (d  + 2) →  )
    (k' h' : Fin d' →  ) (    : (Fin (d  + 2) →  ) → (Fin d' →  ) →  ) :   :=
   v, blockDivisorDensity   p (toNatFun k 1) (d  + 1) k' h'
    (fun v =>   0 v) (fun v =>   0 v) v  (boxMeasure b d')
theorem blockStateIntegral_tendsto ... (hp :   j, finExp k h j = p)
    (hq :   i, p < ...) ... :
    Tendsto (fun N :   => N ^ p / Real.log N ^ (d  + 1)
      * blockStateIntegral   b N k h k' h'    ) atTop (  (blockCoeff   b p k k' h'    ))
theorem blockStateIntegral_isEquivalent ... (h pos : ...) :
    (fun N :   => blockStateIntegral   b N k h k' h'    ) ~[atTop]
      fun N :   => blockCoeff   b p k k' h'     * (N ^ (-p) * Real.log N ^ (d  + 1))
theorem blockStateIntegral_sqrt_isEquivalent ... :
    (fun n :   => blockStateIntegral   b (Real.sqrt n) k h k' h'    ) ~[atTop]
      fun n :   => blockCoeff   b p k k' h'     / 2 ^ (d  + 1)
        * (n ^ (-(p / 2)) * Real.log n ^ (d  + 1))
```

2 Proof strategy

$N^p Z/\log^{d_0+1} N = N^p(Z - Z_0)/\log^{d_0+1} N + N^p Z_0/\log^{d_0+1} N$; the first term tends to 0 (unit 78) and the second to the block divisor coefficient (unit 75, through the identification of the frozen block state integral with the frozen state integral). Positivity of the coefficient is `integral_pos_iff_support_of_nonneg_ae` with the open box inside the support, as in unit 71; integrability is domination by $K \prod_i v_i^{h_i'} (v_i^{k_i'})^{-p}$ using monotonicity of A_{p-1} in a. The equivalence is `isEquivalent_const_iff_tendsto` multiplied by the reference function; the n-form composes with $\sqrt{n} \to \infty$ and uses $\log \sqrt{n} = \frac{1}{2} \log n$.

3 Roadblocks and resolutions

- After `field_simp` the shape of the target no longer matches the intended `linear_combination`; instead rewrite the product as $(N^{-p} N^{p})(\log^{d} N / \log^{d} N)\, Z$ by `ring` and close with the two cancellation facts.

4 Outcome

The Astra plan (targets 1–5) is complete: the leading term of the Taylor-tree theorem is formalised for general continuous ξ, η at every multiplicity. Remaining for the leading term: the Fubini bridge from the iterated block/noncritical form to the genuine box over $\mathrm{Fin}(m + d')$ with an arbitrary coordinate order, and the free-energy corollary. Beyond: the quantitative remainder (Astra target 6) and the state-density expansion transfer (targets 7–8).